EPS Rapid-Review Cheat Sheet (Module 4)

E1 - SVT mechanisms

- **Two engines:** re-entry (a loop; needs 2 paths + unidirectional block + slow conduction) vs automaticity (a lone focus). Triggered activity = afterdepolarizations.
- **RP interval decides the SVT:** SHORT RP = typical AVNRT (slow-fast, P buried) or orthodromic AVRT. LONG RP = "PAT is Late" = PJRT, Atypical AVNRT (fast-slow), atrial Tachycardia.
- **AVNRT** = re-entry inside the AV node; commonest regular SVT; young woman, neck pounding (cannon A waves).
- **AVRT (needs accessory pathway):** orthodromic = down AVN (NARROW), up accessory (retrograde limb), common; antidromic = down accessory (WIDE, VT-like), rare.
- **Delta wave = accessory conducting DOWN** (baseline sinus only; disappears in orthodromic tachy). Concealed pathway = no delta but still AVRT.
- Focal AT: most common LEFT site = pulmonary vein; dig toxicity = "PAT with block". Adenosine terminates AVNRT/AVRT, NOT atrial tachycardia.
- Mahaim = right-sided decremental atriofascicular pathway (antidromic, LBBB). PJRT = slow concealed posteroseptal pathway, incessant, kids, long RP.

E2 - Accessory pathway localization + WPW SCD risk

- Localize on the SINUS ECG (delta present). Order: **V1 (side) -> inferior leads (level) -> lead I (lateral).**
- **V1 UP = LEFT-sided; V1 DOWN = RIGHT. Inferior (II/III/aVF) NEGATIVE = posteroseptal. Lead I/aVL NEGATIVE = left lateral.**
- Concealed pathway can't be localized (no delta). Septal pathways sit near His -> AV-block risk at ablation.
- **WPW SCD risk:** HIGH if **SPERRI ≤ 250 ms in AF** (fast pathway). LOW if intermittent pre-excitation, sudden loss on exercise, or decremental conduction (weak pathway).

E3 - Pacing

- **NBG code:** 1 Paced / 2 Sensed / 3 Response (I inhibit, T trigger, D both) / 4 Rate (R sensor) / 5 Multisite (V = CRT only, else dropped).
- **Mode by problem:** SSS -> **DDDR** (chronotropic incompetence + future AV-block risk). AV block -> DDD. AF + slow ventricle -> VVIR.
- Pace narrow if atrium paced (own His); wide LBBB-like if RV paced. Upper rate only exists with a sensor OR atrial tracking (VVI fixed = none).
- **Alternating RBBB/LBBB = Class I** pacing even if asymptomatic. Reversible bradycardia (jaundice, drugs, thyroid), young vasovagal, and early post-op block = do NOT pace.
- **Congenital complete heart block:** pace only when a red flag appears (symptoms, wide/low escape, LV dysfunction, complex ectopy, pauses > 2-3x cycle, mean HR too low, long QT). Short-coupled PVCs = NOT an indication. ⚠ Q13: key says D, clinical answer is B.

E4 - CRT (biventricular pacing)

- Fixes DYSSYNCHRONY, not rate. LBBB -> lateral LV wall activated last -> pre-excite it with the LV lead.
- **Indication:** EF ≤ 35%, sinus, LBBB, QRS ≥ 150 (strong) / 130-149 (weaker), NYHA II-IV, on optimal meds. QRS < 130 = no benefit / harm.

- 3 leads: RA, RV, LV via coronary sinus into the **posterolateral vein** (the late wall). Need > 98% biventricular capture; AF/PVCs steal it (native beats inhibit the device) -> rate control or AV-node ablation.
- LV-paced ECG: dominant R in V1 + negative lead I (confirms LV capture). CRT-P over CRT-D if short life / poor renal function (non-shockable deaths, competing mortality) / non-ischaemic.

E5 - VT & ARVC

- **3 mechanisms:** reentry (scar), automaticity (focus), triggered (EAD early = long QT; DAD late = calcium). Regular wide-complex tachy = VT until proven otherwise.
- **RVOT VT** = commonest idiopathic; triggered (cAMP: catecholamine -> cAMP -> Ca -> DAD); LBBB + INFERIOR axis; adenosine-sensitive; benign.
- **Fascicular VT** = reentry (left posterior fascicle); RBBB + LEFT axis; verapamil-sensitive; young. (Adenosine tells them apart: stops RVOT, not fascicular.)
- Digoxin VT = DAD, often BIDIRECTIONAL (alternating QRS axis = dig tox or CPVT). Scar VT = reentry around old MI (needs ICD).
- **ARVC:** fibrofatty RV (desmosomal; exercise tears it), LBBB-morphology VT; epsilon wave + T-inversion V1-V3 + delayed S upstroke. Tx: no sport, beta-blocker (amiodarone if refractory); ICD Class I (aborted VF / unstable VT), IIa (stable VT); ablation for recurrent shocks.

E6 - Channelopathies & SCD

- **QTc:** normal ≤ 440 (M) / ≤ 460 (W); **> 500 = dangerous**; short < 330-340. Hypocalcemia = long QT; hypercalcemia = short QT.
- **Torsades** = polymorphic VT of long QT via EAD; Tx IV **magnesium**, correct K, raise rate, stop drug. Acquired long QT: macrolides, antipsychotics, class Ia/III, low K/Mg/Ca, bradycardia. NOT hypercalcemia (Q10).
- **LQT1** (KCNQ1, IKs; exercise/swim; broad T; beta-blocker best). **LQT2** (KCNH2/HERG, IKr = deactivated K; emotion/startle; notched T). **LQT3** (SCN5A gain, late Na; rest/sleep; long ST + late T; mexiletine).
- **Brugada** = SCN5A LOSS; regional RVOT-epicardium Ito unmasked -> dome collapse -> **coved ST V1-V2** (NOT a QT change); rest/sleep/FEVER + Na-blockers worsen; only type-1 diagnostic; ICD only proven. Gene = SCN5A (Q15).
- **Short QT** = K-channel gain; tall peaked T. **CPVT** = RyR2 Ca leak; NORMAL resting ECG; exercise/emotion -> bidirectional VT; beta-blocker + flecainide + no sport + ICD.
- **HCM Risk-SCD (7 factors:** max wall thickness, LA size, LVOT gradient, age, NSVT, unexplained syncope, FHx SCD). Survived VF / unstable VT = ICD Class I; else 5-yr risk ≥ 6% = IIa, 4-6% = IIb. High-risk SCD except (Q14) = stress-induced PVCs.

E7 - ECG rules (four "EXCEPT" lists, one impostor each)

- **Low voltage** (<5 mm limb / <10 mm chest): effusion, obesity, COPD, massive MI, amyloid/sarcoid/iron, myxoedema, constriction. NOT pregnancy (Q3).
- **Axis (leads I + aVF):** both up = normal; I up/aVF down = LEFT; I down/aVF up = RIGHT. LAD: LAHB, inferior MI, LVH, WPW, RV pacing, hyperK, **primum ASD**. RAD: RVH, **secundum ASD (RSR' V1)**, PE, COPD. ASD trap: secundum = RIGHT, primum = LEFT (Q4 = secundum).
- **Pregnancy:** fast rate, LEFT axis, **PR SHORTENS**, small Q/inverted T in III, T changes V1-3. PR prolongation is the impostor (Q5).
- **Neonate:** right axis + dominant R V1, T-inversion V1-3 (juvenile), short intervals, small septal q. Short QTc ≤ 280 = abnormal (Q6).

E8 - AF & anticoagulation

- **CHA2DS2-VASc:** C1 H1 A2(≥75)=2 D1 S2(stroke/TIA)=2 V1 A(65-74)1 Sc(female)1. Treat men ≥ 2, women ≥ 3 (female counts only with another factor). ⚠ Q19: key 3 vs clinical 4. HAS-BLED flags fixable factors, never withholds OAC.
- **NOAC doses:** rivaroxaban **20 OD** (15 if CrCl 15-49, **renal only**); apixaban **5 BID** (2.5 if **2 of 3**: age ≥80, wt ≤60 kg, Cr ≥1.5) = **best for GI bleed**; dabigatran 150 BID (110 if elderly/verapamil, **judgment**); edoxaban 60 OD (30 if **ANY 1 of**: CrCl 15-50 / wt ≤60 / P-gp inhibitor; **avoid if CrCl >95**). "-xaban" = factor Xa; dabigatran = thrombin.
- **P-gp inhibitor** (verapamil, dronedarone, amiodarone, quinidine, ciclosporin, azoles, macrolides) = blocks the gut efflux pump → more NOAC absorbed → higher level/bleed → cut dose. Dabigatran most P-gp dependent.
- **Anticoagulated stroke reperfusion:** thrombectomy needs no reversal (always open). rtPA barred on a NOAC UNLESS reversed: **dabigatran + idarucizumab** unlocks rtPA (Xa drugs' andexanet is for bleeding, not tPA).
- **Valvular AF (warfarin only):** mechanical valve or moderate-severe mitral stenosis. Everyone else = NOAC.
- **After brain event, restart OAC by 1-3-6-12:** TIA 1, mild 3, moderate 6, severe 12 days. No thrombolysis (rtPA) if INR > 1.7 **(warfarin rule)**. On a **NOAC the INR lies**, so lyse only if last dose >48 h OR drug-specific assay normal (anti-Xa for -xaban; dilute thrombin/ecarin for dabigatran) OR reversed (idarucizumab for dabigatran); else **thrombectomy** (rtPA window 4.5 h < NOAC clearance 48 h).
- **Rhythm control:** normal heart = class Ic (flecainide, propafenone) OK; ischaemic/structural = NO class Ic (Q12 = propafenone), use amiodarone (dronedarone not in HF). Rate control: beta-blocker, diltiazem/verapamil, digoxin.

Confirm with your course: E3 Q13 (key D vs clinical B) and E8 Q19 CHA2DS2-VASc (key 3 vs clinical 4).